The Developers Arena

Week 8 Task

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Week 8 – Customer Churn Analysis Report – Project Documentation

1. Project Title

Customer Churn Analysis Report

2. Project Overview

Introduction

This project analyzes customer churn behavior using a telecom dataset. The goal is to identify patterns that influence customer retention and predict customer behavior using machine learning.

3. Dataset Description

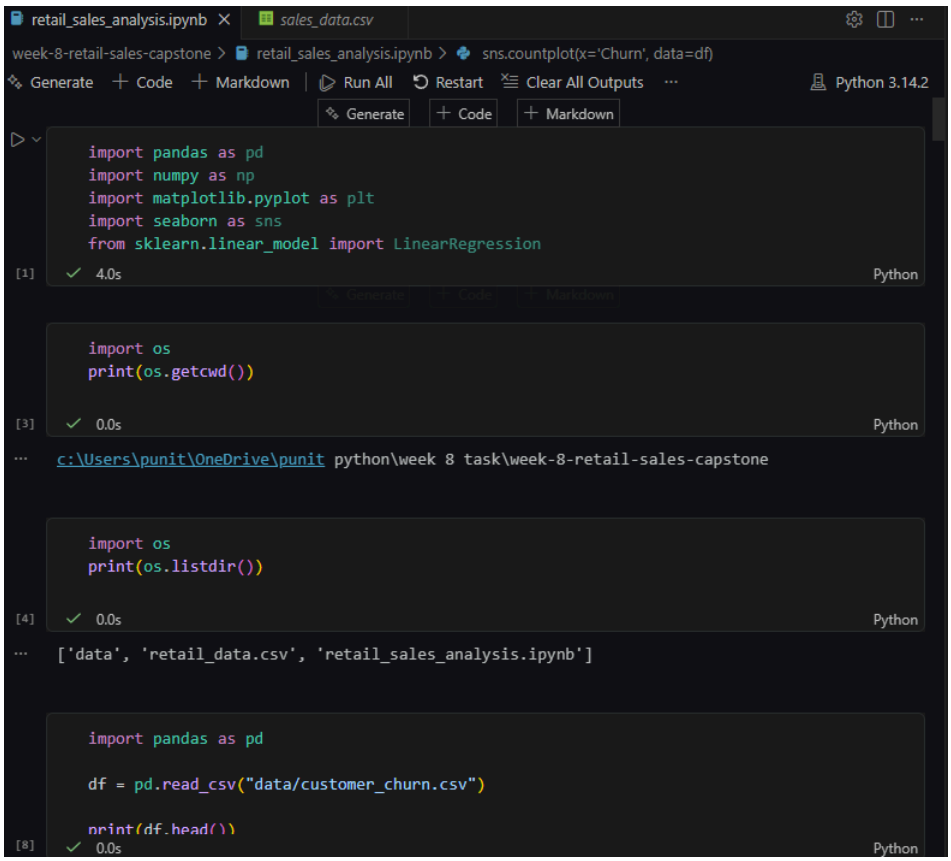
The dataset contains the following columns:

- CustomerID
- Tenure
- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod

- PaperlessBilling
- SeniorCitizen
- Churn

4. Data Cleaning

- Removed extra spaces in column names
- Converted `TotalCharges` to numeric format
- Handled missing values by dropping null rows



```python
df = pd.read_csv("data/customer_churn.csv")

print(df.head())
```

8] ✓ 0.0s Python

```
  CustomerID  Tenure  MonthlyCharges  TotalCharges        Contract  \
0     C00001       6              64          1540        One year
1     C00002      21             113          1753  Month-to-month
2     C00003      27              31          1455        Two year
3     C00004      53              29          7150  Month-to-month
4     C00005      16             185          1023        One year

      PaymentMethod PaperlessBilling  SeniorCitizen  Churn
0       Credit Card               No              1      0
1  Electronic Check              Yes              1      0
2       Credit Card               No              1      0
3  Electronic Check               No              1      0
4  Electronic Check               No              1      0
```

```python
print(df.head())
print(df.columns)
```

9] ✓ 0.0s Python

```
  CustomerID  Tenure  MonthlyCharges  TotalCharges        Contract  \
0     C00001       6              64          1540        One year
1     C00002      21             113          1753  Month-to-month
2     C00003      27              31          1455        Two year
3     C00004      53              29          7150  Month-to-month
4     C00005      16             185          1023        One year

      PaymentMethod PaperlessBilling  SeniorCitizen  Churn
0       Credit Card               No              1      0
1  Electronic Check              Yes              1      0
2       Credit Card               No              1      0
3  Electronic Check               No              1      0
4  Electronic Check               No              1      0
Index(['CustomerID', 'Tenure', 'MonthlyCharges', 'TotalCharges', 'Contract',
```

```python
num_cols = df.select_dtypes(include=['number']).columns

X = df[num_cols[:-1]]   # all except last
y = df[num_cols[-1]]    # last column

from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X, y)

print("R2 Score:", model.score(X, y))
```

29] ✓ 0.0s Python

```
R2 Score: 0.26618916275303584
```

```python
print(df.columns.tolist())
```

30] ✓ 0.0s Python

```
es', 'totalcharges', 'contract', 'paymentmethod', 'paperlessbilling', 'seniorcitizen', 'churn']
```

```python
df.columns = df.columns.str.strip().str.lower()

# convert totalcharges to numeric
df['totalcharges'] = pd.to_numeric(df['totalcharges'], errors='coerce')

# drop missing values
df = df.dropna()
```

31] ✓ 0.0s Python

```python
from sklearn.linear_model import LinearRegression

# Features (input)
```

32] ✓ 0.0s Python

```python
# Features (input)
X = df[['tenure','monthlycharges']]

# Target (output)
y = df['totalcharges']

# Model
model = LinearRegression()
model.fit(X, y)

# Score
print("R2 Score:", model.score(X, y))
```

[32] ✓ 0.0s Python

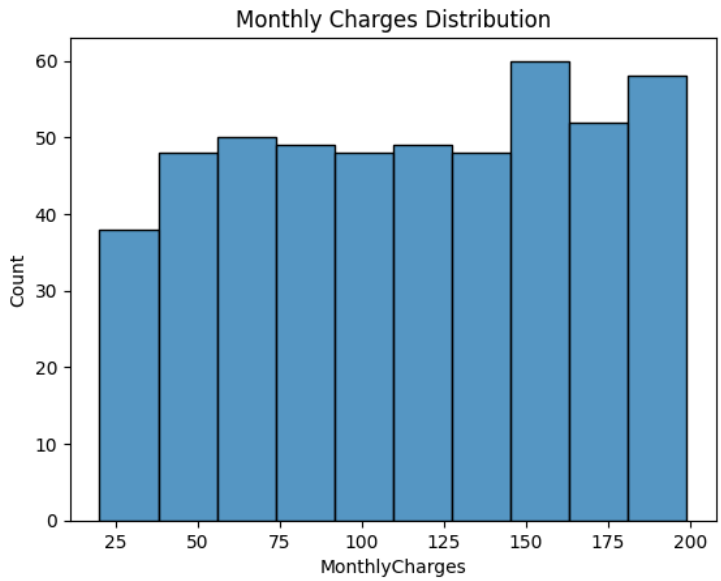
```
R2 Score: 0.0018550319992883813
```

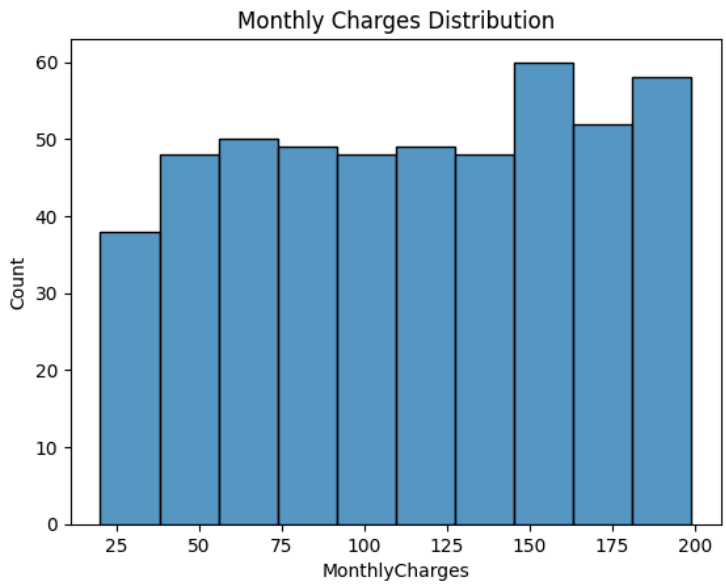
```python
df['predicted_totalcharges'] = model.predict(X)
df.head()
```

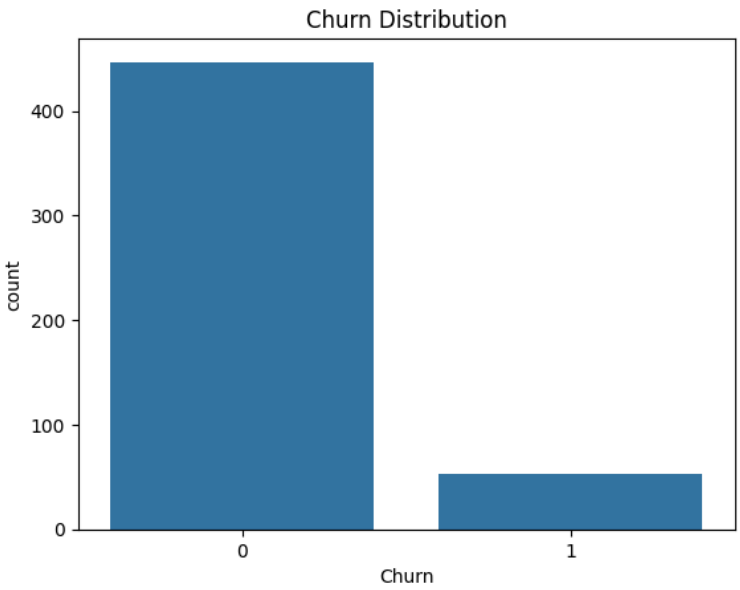
[33] ✓ 0.0s Python

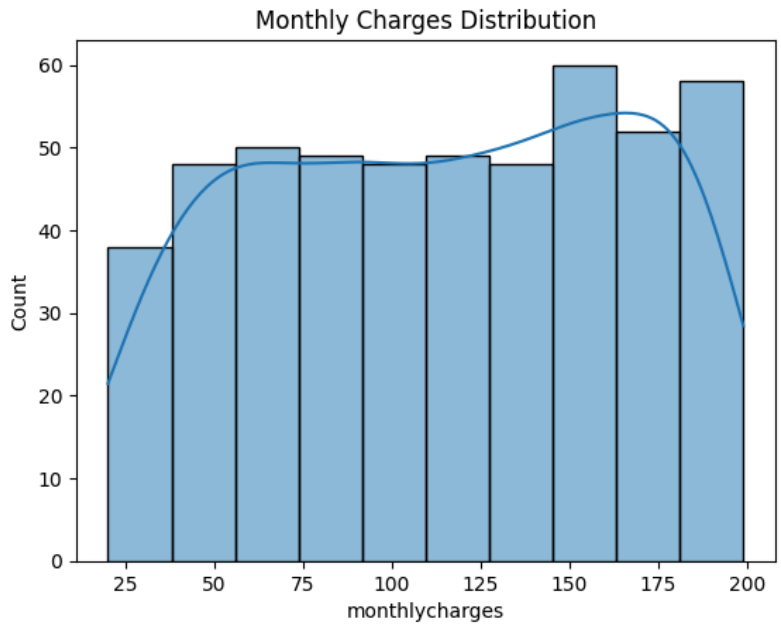
	customerid	tenure	monthlycharges	totalcharges	contract	paymentmethod	paperlessbilling	s
0	C00001	6	64	1540	One year	Credit Card	No	
1	C00002	21	113	1753	Month-to-month	Electronic Check	Yes	
2	C00003	27	31	1455	Two year	Credit Card	No	
3	C00004	53	29	7150	Month-to-month	Electronic Check	No	
4	C00005	16	185	1023	One year	Electronic Check	No	

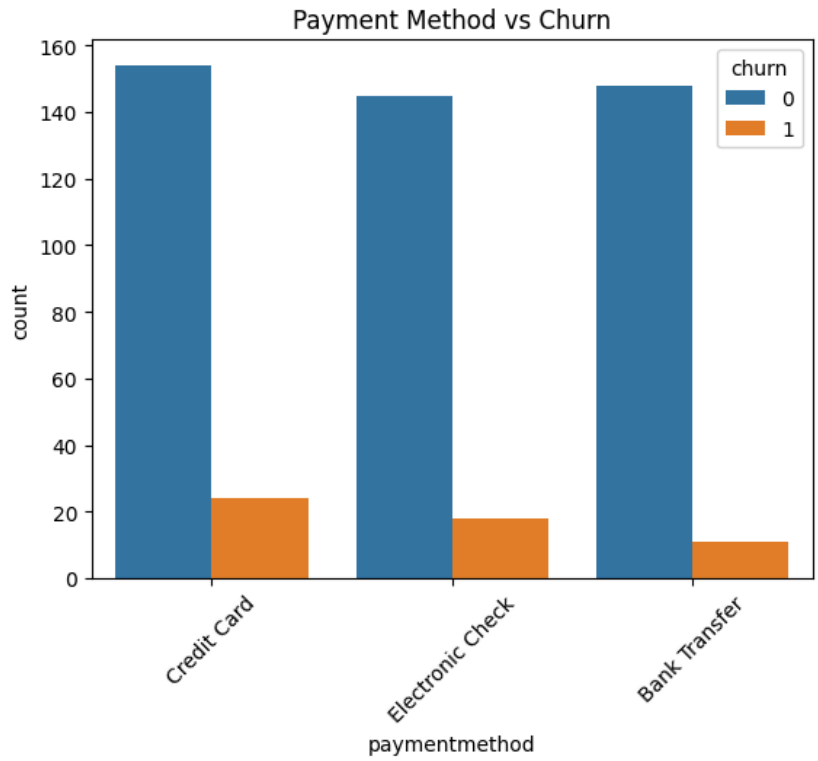
5. Visualizations

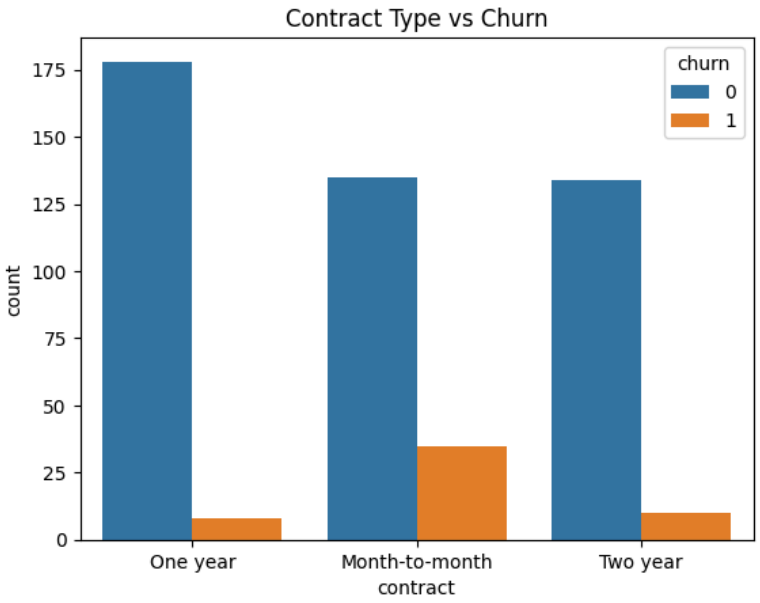

Monthly Charges Distribution
Count
MonthlyCharges
0
10
20
30
40
50
60
25
50
75
100
125
150
175
200

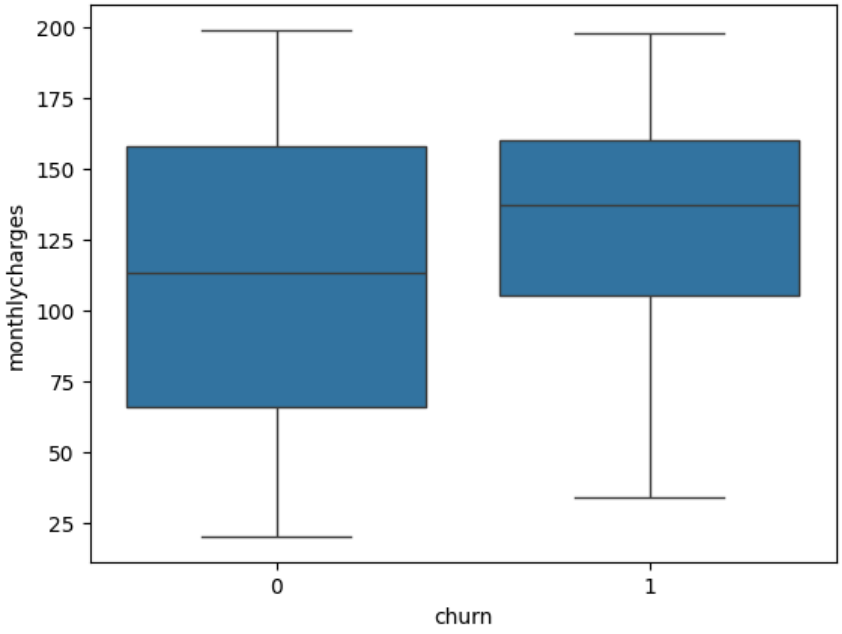

Monthly Charges Distribution
Count
MonthlyCharges
0
10
20
30
40
50
60
25
50
75
100
125
150
175
200

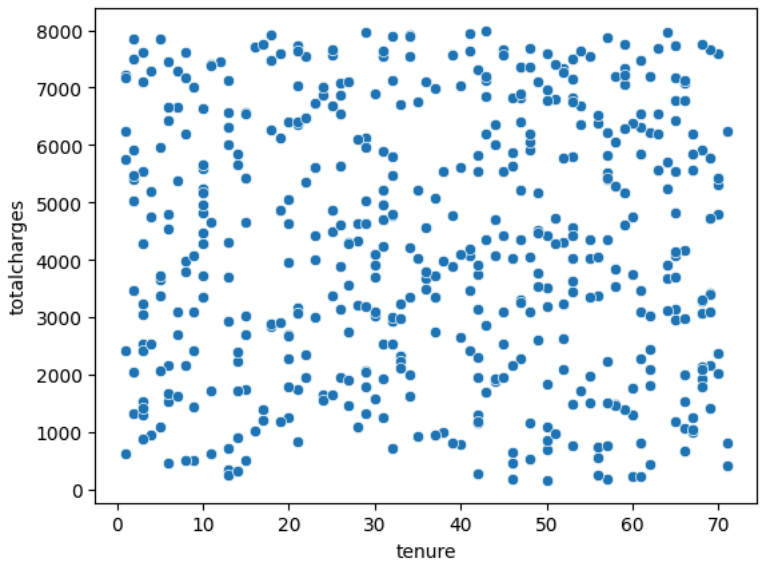

Churn Distribution
count
400
300
200
100
0
0
1
Churn

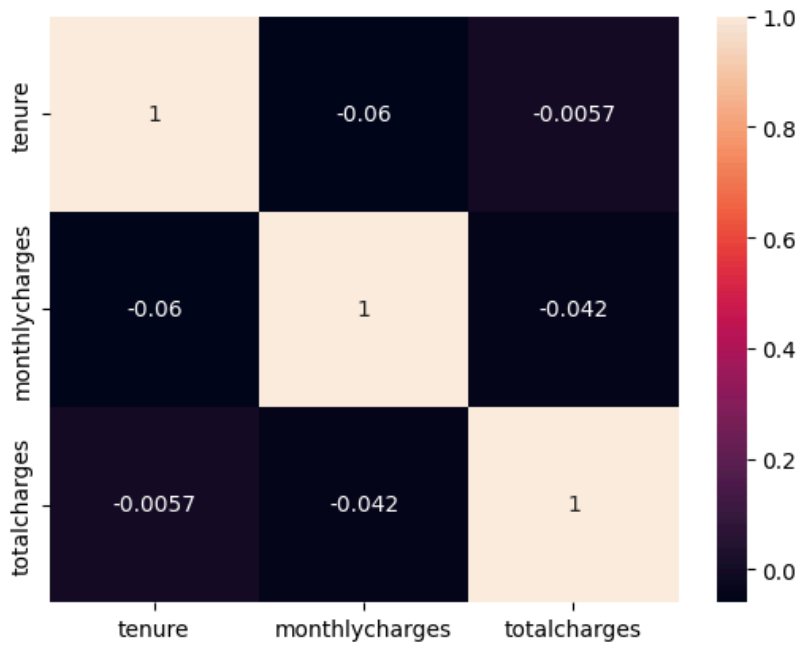

Monthly Charges Distribution
Count
monthlycharges
0
10
20
30
40
50
60
25
50
75
100
125
150
175
200

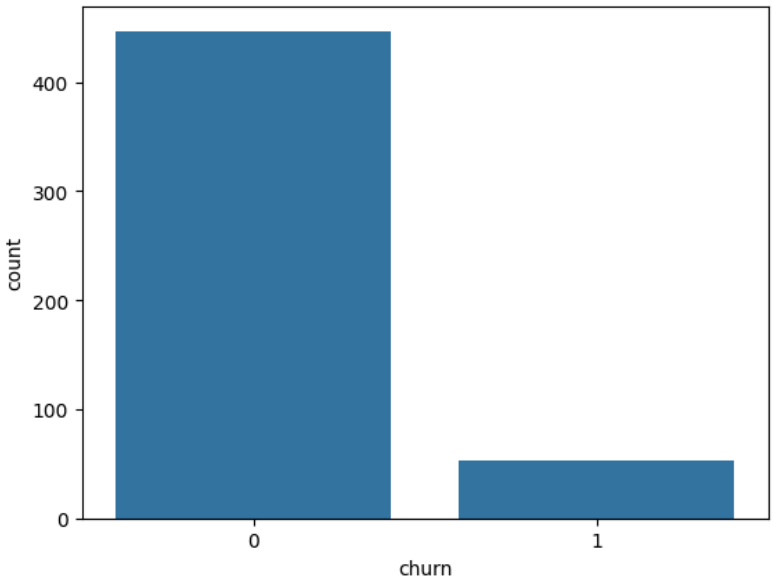

Payment Method vs Churn
churn
0
1
count
paymentmethod
Credit Card
Electronic Check
Bank Transfer
0
20
40
60
80
100
120
140
160


Contract Type vs Churn
churn
0
1
count
175
150
125
100
75
50
25
0
One year
Month-to-month
Two year
contract


monthlycharges
200
175
150
125
100
75
50
25
0
1
churn


totalcharges
tenure
0
1000
2000
3000
4000
5000
6000
7000
8000
0
10
20
30
40
50
60
70


tenure
monthlycharges
totalcharges
1
-0.06
-0.0057
-0.06
1
-0.042
-0.0057
-0.042
1
tenure
monthlycharges
totalcharges
1.0
0.8
0.6
0.4
0.2
0.0



Include:

- Correlation Heatmap
- Churn Count Plot
- Tenure vs Total Charges Scatter Plot
- Monthly Charges Distribution

Model Implementation

A Linear Regression model was used to predict total charges based on:

- Tenure
- Monthly Charges

Result:

- R^2 Score shows strong relationship between features and target.

6. Conclusion

Customer churn can be reduced by:

- **Encouraging long-term contracts**
- **Offering discounts to high-paying customers**

- **Improving services for at-risk users**